Forgetting a Simple Root of a Binary Cubic

A coordinate-free model for a noninjective Keller map in dimension three

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Abstract

We describe intrinsically a polynomial map $F\colon \mathbb{A}^3_{\mathbb{C}} \to \mathbb{A}^3_{\mathbb{C}}$ with constant Jacobian determinant -2 and geometric degree three. Let W be a two-dimensional complex vector space, let H be a tangential affine slice of the space of binary cubics, and mark one simple projective root of a cubic in H. The resulting simple-root incidence variety is, exceptionally, isomorphic to $\mathbb{A}^3$. Under this identification, F is the morphism which forgets the marked root. Resultant normalization gives a coordinate-free proof of etaleness; the three roots of a general cubic explain the generic degree. We prove the affine-space identification, recover the explicit polynomial formula, and interpret the discriminant as the precise locus of nonproperness.

Provenance. The explicit coordinate map studied here was publicly announced by Levent Alpöge in an X post of 20 July 2026 [1]; that post credits Akhil for asking about the problem and Fable for working during the World Cup final. Later that day Andy Jiang (@davikrehalt) posted the projective symmetric-product formulation [2], explicitly presenting it in the post as GPT output. Aaron Lou subsequently gave a factorization–resultant derivation and a global affine chart [3]. The purpose of this note is to make the equivalence of these presentations precise and to supply a detailed intrinsic proof.

1 The intrinsic construction

Throughout, varieties are over $\mathbb{C}$. Let W be a two-dimensional vector space and fix a nonzero alternating form $\omega \in \bigwedge^2 W^\vee$. Choose a tensor $\tau \in \operatorname{Sym}^3 W$ of multiplicity type $(2,1)$: after extension to a basis it has the form $u^2 v$ with u, v independent. Equivalently, $[\tau]$ lies on the tangent developable of the Veronese cubic

$$\nu_3\colon \mathbb{P}(W) \longrightarrow \mathbb{P}(\operatorname{Sym}^3 W)$$

but not on the Veronese curve itself. Rescale τ once and for all, and put

$$H_\tau = \left\{ C \in \operatorname{Sym}^3 W^\vee : \langle \tau, C \rangle = 1 \right\}.$$

This is an affine hyperplane in a four-dimensional vector space, hence $H_\tau \simeq \mathbb{A}^3$. Consider the open simple-root incidence

$$X_\tau = \{ (C, [r]) \in H_\tau \times \mathbb{P}(W) : C(r) = 0 \text{ and } \mathrm{d}C_r \neq 0 \}.$$

Our main statement is the following.

Theorem 1.1 (Marked-root model)**.** *The variety X_τ is isomorphic to $\mathbb{A}^3$. Under this identification, the forgetful morphism*

$$\pi\colon X_\tau \longrightarrow H_\tau, \qquad (C,[r]) \longmapsto C,$$

is an etale polynomial map of affine three-spaces, generically of degree three. Its fiber over C is naturally the reduced set of simple projective roots of C.

Remark 1.2. The construction is coordinate-free but not choice-free. Its intrinsic data are (W,ω,τ) together with the chosen nonzero affine levels. The identifications of X_τ and H_τ with $\mathbb{A}^3$ are not canonical. Accordingly, the statement that π is etale is intrinsic, whereas the numerical value -2 of its Jacobian depends on the chosen affine volume coordinates.

1.1 The projective symmetric-product model

The construction above is the affine normalization of the projective picture posted by Jiang [2]. Identify effective degree-three divisors on $\mathbb{P}(W)$ with projective binary cubics,

$$\mathrm{Sym}^3\big(\mathbb{P}(W)\big) \simeq \mathbb{P}\big(\mathrm{Sym}^3 W^\vee\big),$$

and let

$$\mu\colon \mathbb{P}(W)\times \mathrm{Sym}^2\big(\mathbb{P}(W)\big) \longrightarrow \mathrm{Sym}^3\big(\mathbb{P}(W)\big), \qquad (p,D)\longmapsto p+D$$

forget which root was marked. Write R for its ramification divisor and

$$H_\infty = \{[C] : \langle \tau, C\rangle = 0\}.$$

The small diagonal $\{3p : p\in\mathbb{P}(W)\}$ is a twisted cubic. Because τ has type $(2,1)$, the restriction of the defining functional of H_∞ to this twisted cubic has zero divisor $2p+q$ with $p\neq q$. Thus H_∞ is tangent but not osculating to the small diagonal. Conversely, every such hyperplane arises this way and all of them are equivalent under $\mathrm{PGL}(W)$.

There are natural isomorphisms

$$\mathrm{Sym}^3\big(\mathbb{P}(W)\big)\setminus H_\infty \xrightarrow{\sim} H_\tau, \qquad [C]\longmapsto \frac{C}{\langle \tau, C\rangle},$$

and

$$\Big(\mathbb{P}(W)\times \mathrm{Sym}^2\big(\mathbb{P}(W)\big)\Big)\setminus\Big(R\cup \mu^{-1}(H_\infty)\Big) \xrightarrow{\sim} X_\tau.$$

Under these isomorphisms, μ becomes π. Indeed, deleting R says exactly that the marked root is simple; the two unmarked roots are still allowed to coincide. Lemma 2.1 below gives a further affine lift: writing the marked cubic as $C=\ell q$ and imposing $\mathrm{Res}_\omega(\ell,q)=1$ fixes the remaining factor-rescaling gauge. Hence Jiang's projective restriction, the normalized factor space, and the marked-root incidence are isomorphic presentations of one morphism. The identifications are not literal equalities of coordinate formulas.

The word "exceptionally" belongs to the assertion $X_\tau\simeq\mathbb{A}^3$. The target is visibly affine three-space, but the same need not be expected of a simple-root incidence for an arbitrary slice. The type-$(2,1)$ tensor is the special geometric datum behind the construction.

2 Resultant gauge and marked roots

The volume form gives an intrinsic scalar normalization of the linear–quadratic resultant. For $\ell \in W^\vee$, let $v_\ell \in W$ be determined by

$$\iota_{v_\ell}\omega = \ell.$$

For $q \in \operatorname{Sym}^2 W^\vee$, define

$$\operatorname{Res}_\omega(\ell, q) = q(v_\ell).$$

In an ω-unimodular basis this is the classical binary resultant of a linear and a quadratic form. In particular, it is nonzero precisely when ℓ and q are coprime. Define the normalized factor space

$$Y_{\omega,\tau} = \left\{(\ell, q) \in W^\vee \times \operatorname{Sym}^2 W^\vee : \operatorname{Res}_\omega(\ell, q) = 1, \quad \langle \tau, \ell q\rangle = 1\right\}. \tag{1}$$

Lemma 2.1 (Resultant gauge)**.** *Multiplication and passage to the root line induce an isomorphism*

$$Y_{\omega,\tau} \xrightarrow{\sim} X_\tau, \qquad (\ell, q) \longmapsto (\ell q, \mathbb{P}(\ker \ell)).$$

Proof. A marked root of C is a factorization $C = \ell q$, unique up to

$$(\ell, q) \longmapsto (s\ell, s^{-1}q), \qquad s \in \mathbb{G}_{\mathrm{m}}.$$

The root is simple if and only if ℓ and q are coprime. Moreover,

$$\operatorname{Res}_\omega(s\ell, s^{-1}q) = s \operatorname{Res}_\omega(\ell, q).$$

Thus every coprime scaling orbit has a unique resultant-one representative. Because the normalizing scalar is the inverse of a regular nowhere-zero function, this construction is algebraic in both directions. □

The weight one in the last formula is decisive: multiplication has a $\mathbb{G}_{\mathrm{m}}$-gauge, and the resultant cuts each coprime orbit exactly once.

There is also a root-gradient formulation. If $C = \ell q$ and $r = v_\ell$, then

$$\mathrm{d}C_r = q(r)\ell = \operatorname{Res}_\omega(\ell, q)\, \iota_r\omega.$$

Consequently $Y_{\omega,\tau}$ is equivalently the space of pairs

$$(C, r) \in H_\tau \times (W \setminus \{0\}), \qquad \mathrm{d}C_r = \iota_r\omega. \tag{2}$$

Euler's identity gives $3C(r) = \mathrm{d}C_r(r) = 0$, while the right-hand side is a nonzero covector. Thus the equation simultaneously says that r is a root and that it is simple. Conversely, every simple projective root has a unique representative satisfying (2).

3 A coordinate-free proof of etaleness

Consider the five-dimensional polynomial map

$$\Psi\colon W^\vee \times \operatorname{Sym}^2 W^\vee \longrightarrow \operatorname{Sym}^3 W^\vee \times \mathbb{A}^1, \qquad (\ell, q) \longmapsto (\ell q, \operatorname{Res}_\omega(\ell, q)).$$

Proposition 3.1. *The morphism Ψ is etale wherever $\operatorname{Res}_\omega(\ell, q) \neq 0$. Consequently, $\pi\colon X_\tau \to H_\tau$ is etale.*

Proof. Suppose that ℓ and q are coprime. If

$$\dot{\ell}q + \ell\dot{q} = 0,$$

then unique factorization implies that ℓ divides $\dot{\ell}$. Degree considerations give $(\dot{\ell}, \dot{q}) = a(\ell, -q)$ for some $a \in \mathbb{C}$. Hence the kernel of the differential of multiplication is exactly the infinitesimal gauge direction.

The resultant has gauge weight one, so

$$\mathrm{d}\operatorname{Res}_\omega\big|_{(\ell,q)}(\ell, -q) = \operatorname{Res}_\omega(\ell, q) \neq 0.$$

Therefore $\mathrm{d}\Psi$ is injective, hence an isomorphism between tangent spaces of dimension five. Finally,

$$Y_{\omega,\tau} = \Psi^{-1}(H_\tau \times \{1\}),$$

and its map to H_τ is the base change of Ψ along $H_\tau \times \{1\}$. Etaleness is preserved by base change, and Lemma 2.1 identifies this map with π. □

In coordinates the same argument is the relative-invariant identity

$$\det \mathrm{d}\big(\operatorname{coeff}(\ell q), \operatorname{Res}_\omega(\ell, q)\big) = \pm \operatorname{Res}_\omega(\ell, q)^2.$$

The differential proof explains the cancellation without expanding a 5×5 determinant.

4 The affine-space miracle

It remains to prove the substantive assertion $X_\tau \simeq \mathbb{A}^3$. Choose an adapted basis T, S of $W^\vee$, normalized so that $\omega = T \wedge S$ and $\langle \tau, C \rangle$ extracts the coefficient of T^2S. Write

$$\ell = xT + \beta S, \qquad q = \gamma T^2 + \delta TS + \epsilon S^2.$$

Then (1) becomes the two equations

$$x\delta + \beta\gamma = 1, \qquad x^2\epsilon - x\beta\delta + \beta^2\gamma = 1. \tag{3}$$

This resultant-normalized affine presentation and the chart below are the factorization construction given by Lou [3]; we include the inverse to make the global affine-space assertion explicit.

Proposition 4.1 (Global affine chart)**.** *The map $\chi\colon \mathbb{A}^3_{x,y,z} \to Y_{\omega,\tau}$ defined by*

$$\begin{aligned}
\beta &= 1 + xy, \\
\gamma &= \frac{2 - 3xy + x^2 z}{2}, \\
\delta &= -\frac{xz - y + x^2 yz - 3xy^2}{2}, \\
\epsilon &= -z(1 + xy)^2 + 4y^2 + 3xy^3
\end{aligned}$$

is an isomorphism. Its polynomial inverse is

$$x = x, \qquad y = 2\beta\delta - x\epsilon, \qquad z = 4y\delta + 2\gamma y^2 - \epsilon. \tag{4}$$

Proof. Substitution verifies both equations in (3). Conversely, the two equations of the slice imply

$$\beta - 1 = x(2\beta\delta - x\epsilon) = xy,$$

so $\beta = 1 + xy$. Substitution into the first slice equation, followed by the definition of z in (4), gives

$$2\gamma = 2 - 3xy + x^2 z$$

and then

$$2\delta = y(1 + 3xy) - xz(1 + xy).$$

Finally the formula defining z gives

$$\epsilon = -z(1 + xy)^2 + y^2(4 + 3xy).$$

These are precisely the displayed coordinate functions. Both compositions are therefore the identity, including on $x = 0$; no divisor has been removed. □

This completes the proof that both source and target of π are affine three-spaces. Together with Proposition 3.1, it proves the etaleness assertion of Theorem 1.1.

5 The explicit Keller map

Write a cubic in H_τ as

$$C = c_3 T^3 + T^2 S + c_1 T S^2 + c_0 S^3$$

and use affine target coordinates $(2c_3, 2c_1, c_0)$. Multiplication $C = \ell q$, followed by the chart of Proposition 4.1, yields $G = (G_1, G_2, G_3)$ with

$$\begin{aligned} G_1 &= x^3 z - 3x^2 y + 2x, \\ G_2 &= -3x^3 y^2 z + 9x^2 y^3 - 6x^2 yz + 12xy^2 - 3xz + y, \\ G_3 &= -x^3 y^3 z + 3x^2 y^4 - 3x^2 y^2 z + 7xy^3 - 3xyz + 4y^2 - z. \end{aligned}$$

Because G is etale, its Jacobian determinant is a unit of $\mathbb{C}[x, y, z]$, hence a nonzero constant. Its linear part at the origin is

$$(x, y, z) \longmapsto (2x, y, -z),$$

and therefore

$$\boxed{\det JG = -2.}$$

The map announced by Alpöge [1], in the original coordinate order, is

$$\begin{aligned} F_1 &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

Indeed, if $S_0(x, y, z) = (x, y, -z)$ and $R_0(a, b, c) = (c, b, a)$, then

$$G = R_0 \circ F \circ S_0.$$

Thus F and G are exactly left–right linearly equivalent presentations of the same etale morphism. This is an isomorphism of source–target arrows, not strict dynamical conjugacy: the source and target coordinate changes are independent.

There is an equally compact normalized-root certificate. For $F(x, y, z) = (A, B, C)$ define

$$Q_{A,B,C}(U, V) = CU^3 - 2U^2V + BUV^2 - 2AV^3$$

and set $r = (U, V) = (1 + xy, x)$. Directly,

$$Q_{A,B,C}(r) = 0, \qquad (Q_{A,B,C})_U(r) = 2V, \qquad (Q_{A,B,C})_V(r) = -2U.$$

For $\omega_0 = \mathrm{d}U \wedge \mathrm{d}V$, this says

$$\mathrm{d}Q_{A,B,C}\big|_r = -2\,\iota_r\omega_0.$$

It is the same marked-root normalization as (2), after the harmless source and target rescalings relating F and G.

6 Fibers, discriminant, and infinity

The fiber description is immediate from the moduli interpretation.

Proposition 6.1. *For $C \in H_\tau$, the geometric fiber $\pi^{-1}(C)$ is the reduced set of simple projective roots of C. Its cardinality is*

$$\#\pi^{-1}(C) = \begin{cases} 3, & C \text{ has three simple roots}, \\ 1, & C \text{ has a double root and a simple root}, \\ 0, & C \text{ has a triple root}. \end{cases}$$

In particular, π has geometric degree three and is not injective.

For instance, $C_0 = TS(T + S)$ lies in H_τ and has three possible marked linear factors. In the coordinates of G these give

$$G(-1, 2, -8) = G(0, 2, 16) = G(1, -1, -5) = (0, 2, 0).$$

Equivalently, in the original coordinates,

$$\begin{aligned} F\left(0, 0, -\frac{1}{4}\right) &= F\left(1, -\frac{3}{2}, \frac{13}{2}\right) \\ &= F\left(-1, \frac{3}{2}, \frac{13}{2}\right) = \left(-\frac{1}{4}, 0, 0\right). \end{aligned}$$

To see the boundary mechanism, introduce the full incidence

$$\overline{X}_\tau = \{(C, [r]) \in H_\tau \times \mathbb{P}(W) : C(r) = 0\}.$$

Its projection to H_τ is finite flat of degree three. Its ramification locus consists of repeated marked roots, and its branch locus is the binary cubic discriminant. The source X_τ is obtained by deleting precisely that ramification locus upstairs. Over the complement of the discriminant, no point is deleted and π is finite etale. Over a double-root cubic the length-two ramified point is removed while the simple-root point remains; at a triple-root cubic every point is removed.

In the affine chart $X_\tau \simeq \mathbb{A}^3$, the deleted points appear at infinity. Thus π is etale and quasi-finite but not finite or proper. It should not be called a topological three-sheeted covering. Its generic monodromy is the ordinary S_3-monodromy of the roots of a cubic.

For the original target coordinates (a, b, c), a normalized discriminant (one quarter of the standard binary-cubic discriminant) is

$$\Delta(a, b, c) = b^2 - cb^3 - 16a + 18abc - 27a^2c^2.$$

The nonproperness set is exactly $V(\Delta)$. Triple-root cubics form the omitted curve

$$(a, b, c) = \left(\frac{\rho^2}{3}, 2\rho, \frac{2}{3\rho}\right), \qquad \rho \in \mathbb{C}^*.$$

This is the geometric reason the map can be everywhere locally invertible without being globally invertible: its missing inverse branches do not ramify at finite source points; they escape through the boundary of the affine chart.

7 Conclusion

The coordinate formula conceals a short moduli problem:

binary cubic in a tangential affine slice + one normalized simple root $\longmapsto$ the cubic.

The resultant-one condition removes the factor-rescaling gauge, and its weight one proves etaleness. A general cubic has three roots, which explains the degree. The exceptional global chart $X_\tau \simeq \mathbb{A}^3$ converts this forgetful morphism into a polynomial self-map of affine three-space. The discriminant records exactly where inverse branches leave that chart. In this form, constant Jacobian, noninjectivity, and nonproperness are three aspects of the same marked-root geometry.

References

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